

Majid Daliri

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Education	New York University, New York, USA Ph.D. in Computer Science (in progress) Student in the Theoretical Computer Science group Advised by Prof. Christopher Musco	2022 - present
	New York University, New York, USA M.S. in Computer Science	2022 - 2024
	University of Tehran, Tehran, Iran B.S. in Computer Engineering (GPA: 3.97/4.0)	2017 - 2022
Publications	(1) Singular Value Perturbation: Unlocking RL Performance via Diverse Latent Reasoning (Arxiv) Jihwan Shin, Majid Daliri, Vahab Mirrokni, Amir Zandieh, Insu Han (2) On Worst-Case Guarantees for Graph-Based Nearest-Neighbor Search (Neurips) 2026 Majid Daliri, Mohammadreza Daneshvaramoli, Purna Dutta, Edo Liberty, Cameron N Musco, Christopher Musco, Torsten Suel (3) Quantizing Sampling-Based Inner Product Sketches (VecDB@VLDB) 2026 Gabriel M. Silva, Aécio Santos, Majid Daliri, Juliana Freire, Christopher Musco (4) MosaicJoin: Compact Semantic Sketches for Value-Level Join Discovery (VLDB) 2026 Grace Fan, Eden Wu, Majid Daliri, Juliana Freire (5) RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows (Arxiv) H. Mahdavi, P. Mahdavinia, S. Malek, P. Mohammadipour, A. Hashemi, Majid Daliri, A. Farhadi, A. Khasahmadi, N. Miresghallah, V. Honavar (6) CombiGraph-Vis: A Curated Multimodal Olympiad Benchmark for Discrete Math Reasoning (Arxiv) H. Mahdavi, P. Mahdavinia, A. Farhadi, P. Mohammadipour, S. Malek, Majid Daliri, A. Hashemi, A. Khasahmadi, V. Honavar (7) Atlas: Learning to Optimally Memorize the Context at Test Time with Omega Rule (ICML) 2026 Ali Behrouz, Zeman Li, Praneeth Kacham, Majid Daliri, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, Vahab Mirrokni (8) TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate (ICLR) 2026 Amir Zandieh, Majid Daliri, Majid Hadian, Vahab Mirrokni (9) Brains vs. Bytes: Evaluating LLM Proficiency in Olympiad Mathematics (COLM) 2025 H. Mahdavi, A. Hashemi, Majid Daliri, P. Mohammadipour, A. Farhadi, S. Malek, Y. Yazdanifard, A. Khasahmadi, V. Honavar (10) Coupling without Communication and Drafter-Invariant Speculative Decoding (ISIT) 2025 Majid Daliri, Christopher Musco, Ananda Theertha Suresh (11) Unlocking the Theory Behind Scaling 1-Bit Neural Networks (CPAL) 2025 Majid Daliri, Zhao Song, Chiwun Yang (12) QJL: 1-Bit Quantized JL transform for KV Cache Quantization (AAAI) 2025 Amir Zandieh, Majid Daliri, Insu Han (13) Matrix Product Sketching via Coordinated Sampling (ICLR) 2025 Majid Daliri, Juliana Freire, Danrong Li, Christopher Musco (14) Sampling Methods for Inner Product Sketching (VLDB) 2024 Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang (15) Simple Analysis of Priority Sampling (SOSA) 2024 Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang	

- (16) **KDEformer: Accelerating Transformers via Kernel Density Estimation** (ICML) 2023
Amir Zandieh, Insu Han*, **Majid Daliri***, Amin Karbasi (* equal contribution)
- (17) **Weighted Minwise Hashing Beats Linear Sketching for Inner Product Estimation** (PODS) 2023
Aline Bessa, **Majid Daliri**, Juliana Freire, Cameron Musco, Christopher Musco, Aécio Santos, Haoxiang Zhang
- (18) **Efficient Approximations for Cache-conscious Data Placement** (PLDI) 2022
Ali Ahmadi, **Majid Daliri**, Amir Kafshdar Goharshady, Andreas Pavlogiannis
- (19) **A 10-Approximation of the $\frac{\pi}{2}$ -MST** (STACS) 2022
Ahmad Biniaz, **Majid Daliri**, AmirHossein Moradpour

Research Internship

Research Intern, Google, Advised by [Dr. Jieming Mao](#) Summer 2026

- Developing better **agentic AI systems** for Antigravity, focusing on reliable reasoning, tool use, and workflow completion.
- Creating finance-domain datasets and benchmarks for evaluating agents on realistic financial tasks.
- Analyzing agent performance across accuracy, robustness, and end-to-end task success.

Student Researcher, Google, Advised by [Dr. Rajesh Jayaram](#) Spring 2026

- Developed **SliceTurbo**, a sliced variant of TurboQuant designed to accelerate similarity-search pipelines through efficient quantized filtering.
- Studied SliceTurbo across three applications: **nearest-neighbor search acceleration, k-means acceleration, and multi-vector search.**
- Built benchmarks and ablation studies evaluating recall, clustering quality, latency, and memory tradeoffs across retrieval workloads.

Research Scientist Intern, Adobe, Advised by [Dr. Tung Mai](#) Summer 2025

- Developed quantization methods for on-device models under tight memory and compute budgets.
- Built QAT workflows with ONNX and TorchScript for efficient mobile/edge deployment.
- Achieved major model compression and speedups with minimal accuracy drop.

Machine Learning Intern, Max Planck Institute, Advised by [Dr. Amir Zandieh](#) 2021-2022

- Implemented Fast Attention in Transformers, optimizing accuracy and efficiency in sequence modeling tasks.
- Designed an efficient GPU-compatible LSH method, boosting performance in attention approximation.
- Technical Stack: BigGAN, PyTorch, Transformer architectures, and advanced sequence modeling tools.

Research Intern, HKUST, Advised by [Prof. Amir Goharshady](#) 2021-2022

- Made a pioneering theoretical contribution to Cache-conscious Data Placement (CDP), addressing a longstanding challenge in optimizing cache hits.
- Implemented and tested various cache management policies and algorithms, outperforming previous heuristics.
- Our method emerged as the most effective solution, setting a new standard for cache optimization.

Research Intern, University of Salzburg, Advised by [Prof. Ana Sokolova](#) Summer 2022

- Developed algorithms for Distribution Bisimilarity, focusing on finite bisimulation up to convex hull witness.
- Extended research to probabilistic system verification and collaborated on using Quantum Annealers for program verification.

Awards and Honors

Apple Scholars in AI/ML 2025-2027

Full graduate fellowship for the 2025–2027 academic years.

Travel Grants 2024, 2025

Awarded travel grants to present papers at:

- AAAI Conference on Artificial Intelligence (2025)
- ACM-SIAM Symposium on Discrete Algorithms (2024)
- ICERM Fusing Theory and Practice of Graph Algorithms (2025)

Research Grant, University of Salzburg	Summer 2022
Awarded a €5,000 grant for a research internship focusing on algorithms for distribution bisimilarity, probabilistic systems verification, and quantum annealing projects.	
Hong Kong PhD Fellowship Scheme (declined to attend NYU)	2022
Awarded \$185,000 Ph.D. Fellowship as one of the top 300 students selected across all fields for academic excellence and research potential.	
ACM ICPC - Regional (University of Tehran)	2019
Ranked 6 th among more than 100 team all around the Iran.	
Iranian National Olympiad in Informatics Finalist (IOI, Iran)	2016
Awarded to 50 students after a year long competition involving 10,000 students.	

Service

- External Reviewer for [Symposium on Foundations of Computer Science \(FOCS 2025\)](#)
- Reviewer for [International Conference on Learning Representations \(ICLR 2025-2026\)](#)
- Reviewer for [Conference on Neural Information Processing Systems \(NeurIPS 2024-2025\)](#)
- Reviewer for [ACL Rolling Review \(ACL, EMNLP, NAACL 2024-2025\)](#)
- Reviewer for [International Conference on Machine Learning \(ICML 2024-2026\)](#)
- Reviewer for [Royal Society Open](#)
- External Reviewer for [Canadian Conference on Computational Geometry \(CCCG 2023\)](#)

Conference Presentations

- QJL: 1-Bit Quantized JL transform for KV Cache Quantization { Poster, AAAI 2025
Presentation, ICERM 2025
- Coupling without Communication { Presentation, ISIT 2025
Presentation, (NYC TCS Day) 2024
- Simple Analysis of Priority Sampling Presentation, (SOSA) 2024
- Accelerating Transformers via Kernel Density Estimation Poster, (ICML) 2023
- Weighted MinHash for Inner Product Estimation Poster, (PODS) 2023
- Efficient Approximations for Cache-conscious Data Placement Presentation, (PLDI) 2022

Teaching

- Section Leader for [NYU CSCI-UA 310 Basic Algorithms](#) Spring 2023
- Teaching Assistant [NYU CS-GY 6763 Algorithmic Machine Learning](#) Fall 2022
- Teaching Assistant University of Tehran Design and Analysis of Algorithms Fall 2020-2021

Work Experience

- Site Reliability Engineer at [Cafebazaar](#) 2021 - 2022
- Designed, implemented, and maintained both Redis-as-a-Service/PostgreSQL-as-a-Service on a Kubernetes-based cloud.
- Achieved consistent performance benchmarks for both services with 100% uptime and a 99.9% response rate.
- Technical Stack: Kubernetes, Docker, Sentry, S3, Prometheus

Skills

Theoretical Background:

Proficient in Machine Learning Theory, Neural Networks, Linear Algebra, and Probability.

Technical Skills:

Highly skilled in C/C++, CUDA, Go, Python, Bash-Scripting, PHP, JavaScript. Experience with PyTorch, TensorFlow, Django, CSS3, HTML5, and git.